

=====LESSON 9 - Business Data Presented and Analyzed=====

What is Statistics?

- Statistics is a **mathematical body of science** which deals with **collecting, organizing, presenting, analyzing, and interpreting data**.

Frequency Distribution and Graphs

- A tabular or a graphical representation **summarizing the observations in the data** and their corresponding frequencies. **Frequency** refers to the **number of times an observation appears** in the data.

SAMPLE DATA

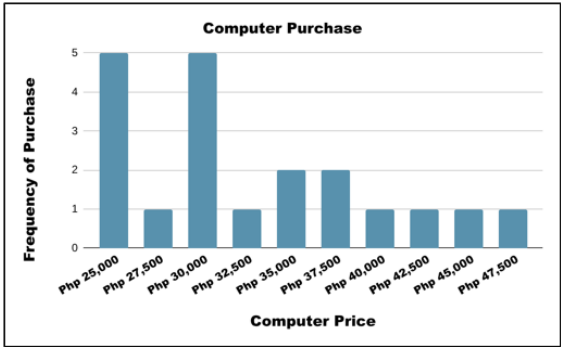
A computer industry marketing agent wants to find out how much money college freshmen are willing to spend to set up a computer in their dormitory rooms. After visiting a college dormitory, the agent gathered the following data on the amount of money 20 students spent on computers.

Php 25,000	Php 37,500	Php 30,000	Php 25,000
Php 32,500	Php 47,500	Php 25,000	Php 35,000
Php 30,000	Php 30,000	Php 30,000	Php 40,000
Php 35,000	Php 25,000	Php 45,000	Php 37,500
Php 27,500	Php 25,000	Php 30,000	Php 42,500

Price of computer	Tally	Frequency
Php 25,000	III-II	5
Php 27,500	I	1
Php 30,000	III-II	5
Php 32,500	I	1
Php 35,000	II	2
Php 37,500	II	2
Php 40,000	I	1
Php 42,500	I	1
Php 45,000	I	1
Php 47,500	I	1
		N = 20

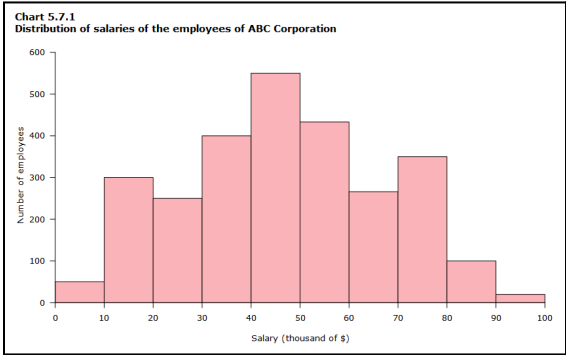
TYPES OF GRAPHS

A. Bar Graph



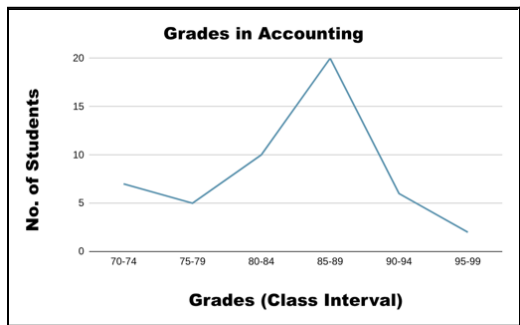
A bar graph is a graphical representation of data using **vertical or horizontal bars or strips**. In a bar graph, the length of the strips is proportional to the numerical value of an observation in the data.

B. Histogram



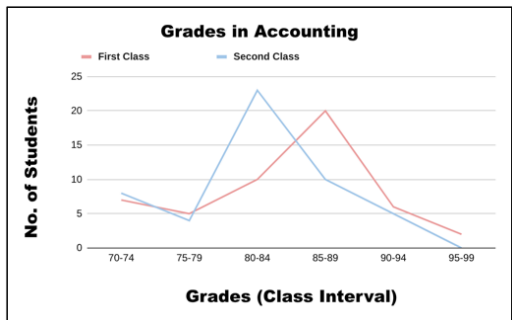
A histogram is the **most common graphical representation of a frequency distribution**. It looks similar to a bar graph but the difference is that each bar represents the number of observations belonging to a specified range of values

C. Line Graph/Frequency Polygon



A line graph is a graphical data representation which graphs the values of data observations **using dots or points connected by lines**.

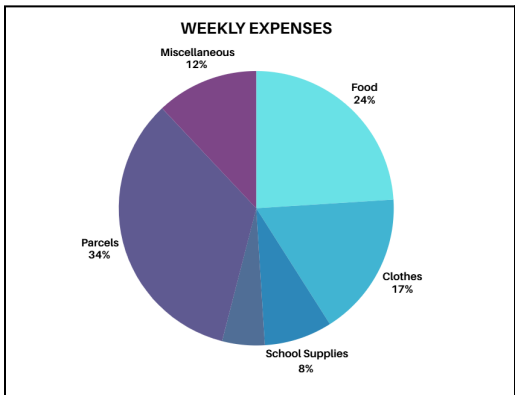
The line graph that will be shown is a **frequency polygon**. This is similar to the histogram, but instead of bars, a straight line segment connects the points on the graph.



A line graph can also be used to **compare two data**.

Assuming the grades of a second Accounting Class were obtained and compared to the grades of the first Account Class.

D. Circle Graphs/ Pie Charts



Circle graphs, often called pie charts, represent data frequencies as pieces or slices of a circular graph.

Constructing a Pie Chart

- Step 1:** Draw a circle
- Step 2:** Take the percentage for each subcategory and convert the percentage decimal.
- Step 3:** Multiply each decimal by 360 degrees represented by the percentage.

Consumer Price Index (CPI)

- measures the weighted average price of a basket of consumer goods and services. Changes in the CPI help assess the cost of living and identify periods of inflation or deflation.

Statistical Table

Consists of **four major parts**:

- a. Title**
The main heading written (Capital letters) on the top of the table.
- b. Box Head or Column Captions**
Caption written as a vertical heading or sub-heading of a column. Only the 1st letter of the heading is in capital letters.
- c. Stub or Row Captions**
Caption written in horizontal heading. These entries at the left side of the table
- d. Body**
Main part of the table which contains the numerical information.

Minor parts:

- a. Prefatory Notes**
A statement below the title enclosed in parenthesis.
- b. Footnotes**
Statements placed immediately below the main part of the table to provide additional information
- c. Source Notes**
Written at the end of the table for the purpose of acknowledging the source of the information
- d. Table Number**
Tables are numbered in the order that they are referred to in the report.

Title PARTS OF A STATISTICAL TABLE TABLE 1: PHILIPPINE PESO RATES																				
Box Head Body Stub	<table> <tr> <th>Top 5 Currencies</th><th>Php 1 equivalent</th><th>1 unit equivalent amount in Peso</th></tr> <tr> <td>EURO</td><td>0.016</td><td>59.32</td></tr> <tr> <td>US \$</td><td>0.0228</td><td>43.82</td></tr> <tr> <td>British Pound</td><td>1.352</td><td>0.74</td></tr> <tr> <td>Indian Rupee</td><td>1.352</td><td>0.74</td></tr> <tr> <td>Australian \$</td><td>0.0243</td><td>41.12</td></tr> </table>	Top 5 Currencies	Php 1 equivalent	1 unit equivalent amount in Peso	EURO	0.016	59.32	US \$	0.0228	43.82	British Pound	1.352	0.74	Indian Rupee	1.352	0.74	Australian \$	0.0243	41.12	Sources: www.x-rates.com/table/?from=php, sponsored by Forex Trading
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General Rules for Tabulation

- 1. A table must be simple for easy computations.
- 2. The numerical information should be arranged in an orderly manner to facilitate valid and convincing conclusions.
- 3. Heading captions must be clear and properly labeled.
- 4. The unit of measurement must be well defined.
- 5. The tables are flexible for possible alterations to meet the changing times.
- 6. If there is a large volume of data, constructing an additional table is suggested.

Group Frequency Table

- organizes a large or wide-ranging dataset into categories called class intervals, showing the frequency (count) of data points falling into each interval.

Consider a large set of data containing scores of students in one subject. The following table summarizes the steps in constructing a grouped frequency distribution of scores.

Constructing a grouped frequency distribution

Step 1: Determine the range. Range is the difference between the highest and lowest values of the data.

Step 2: Approximate the number of classes. Refers to the one group of data. Ideal number of class intervals is from 5 to 10 intervals or group.

Step 3: Determine the class width or class size. The class width is the number of values in one class or group.

Class Width = $\frac{Range}{Number\ of\ Class\ Intervals}$

Step 4: Construct the class intervals with lower and upper limits. The lower limit of the first class may be less than or equal to the lowest value in the data.

To get the lower limit of the next class, add the class size to the current lower limit and do this for all the classes.

The upper limit of each class is obtained by adding one minus the class width from each of the lower limits.

To verify, make sure that each class has the same class width.

Step 5: Tally the number of observations falling within each class interval. The tally in each interval is referred to as frequency.

Step 6: Determine the class mark of each interval. The class mark is the midpoint of each class interval.

Class Mark = $\frac{\text{Lower Limit} + \text{Upper Limit}}{2}$

Step 7: Determine the class boundaries. This is also referred to as the true limits.

Step 7: If the upper and lower limits of the class interval are integers:

lower class boundary (LB) = *lower limit* - 0.5

upper class boundary (UB) = *upper limit* + 0.5

If the data values scores have a decimal value:

lower class boundary (LB) = *lower limit* - 0.05

upper class boundary (UB) = *upper limit* + 0.05

Step 8: Determine the relative frequency. This is the frequency of each class expressed as a percentage of the entire data.

Relative frequency = $(\frac{\text{frequency}}{\text{Total frequency}}) \times 100\%$

Step 9: Determine the less than cumulative frequency of each class. The **cumulative frequency** of the **first class interval** is the **frequency of the first class interval**. The **cumulative frequency of the next class** is obtained by **adding its frequency to the cumulative frequency of the previous class**. To verify, the cumulative frequency of the **last class** must be **equal to the total frequency** or the number of data observations.

SAMPLE PROBLEM

Annie’s bakery bakes and sells cookies. Create a grouped frequency distribution of the number of cookies sold in 30 days from the following data:

200	400	350	280	340	250
150	175	400	275	390	300
190	180	180	145	270	260
360	170	250	220	290	335
400	400	210	300	350	270

=====LESSON 10 - The Measures of Central Tendency=====

Central Tendency

- They are **numerical descriptive measures** that indicate or locate the **center** of the distribution or data set.
- In layman’s terms, a measure of central tendency is an average.
- It is a single number or value that can be considered typical in a set of data as a whole.
- **Mean, Median, Mode**

A. MEAN

- Refers to the **arithmetic average** of a data set.
- A set of values or measurements is the sum of all measurements divided by the number of measurements in the set.
- It is an important descriptive statistic, especially used when the data values are scattered or exhibit no pattern.
- It is also the most reliable measure.
- The mean serves as a good foundation for the computation of advanced statistical methods.

-> Characteristics of the MEAN

- The mean is the most appropriate measure of central tendency when the data are in the interval or ratio scale.
- The mean lies between the largest and the smallest values of measurement.
- There is only one value for the mean for a given set of values or measurements.
- The mean is easily influenced by extreme values because all values contribute to the average. If there are high values, the mean tends to be high also. If there are extremely low values, the mean tends to be below also

a. Arithmetic Mean

- The most widely used numerical measure of the mean or average of the data set.
- It is important to know whether the data at hand represents a population (an entire group) or a sample (a portion or part of a population).

$$\mu = \frac{\Sigma X}{N}$$
$$\bar{x} = \frac{\Sigma x}{n}$$

N = Total number of values in a population

n = Total number of values in a sample

SAMPLE PROBLEM

The owner of the General Merchandise Store wants to know the average daily sales for the week. The accountant has the following data on daily rates.

	Sun	Mon	Tues	Wed	Thurs	Fri	Sat
Daily Rates	₱2,000	₱1,700	₱1,250	₱874.50	₱2,250	₱1,000	₱1,324

$$\bar{x} = \frac{\Sigma x}{n}$$
$$\bar{x} = \frac{10,398.5}{7}$$
$$\bar{x} = 1,485.5$$

b. Weighted Mean

- When the values in the data set appear more than once, businesses often prefer computing the average using the weighted mean.
- A weight, which is expressed as the frequency (f) of each data is multiplied by each value (x).

$$Weighted\ Mean\ (W) = \frac{\Sigma f(x)}{\Sigma f}$$

SAMPLE PROBLEM 1

Consider the following data set containing the daily sales in one week. Compute the weighted mean.

	Sun	Mon	Tues	Wed	Thurs	Fri	Sat
Daily Rates	₱2,000	₱1,000	₱1,000	₱1,600	₱2,000	₱1,600	₱1,600

Sales in Peso (x)	Number of days (f)	Product f(x)
₱2,000	2	₱4,000
₱1,600	3	₱4,800
₱1,000	2	₱2,000
	n = 7	Σf(x) = 10,800

Weighted Mean (W)

=

$\frac{\Sigma f(x)}{\Sigma f}$

Weighted Mean (W)

=

$\frac{10,800}{7}$

Weighted Mean (W)

=

1,542.8571 ≈ 1,542.86

SAMPLE PROBLEM 2

A common application of the concept of weighted mean is when we calculate the Grade Point Average (GPA). Consider the following grades of student A during his 1st semester.

Subjects	Units (f)	Grades (x)	f(x)
ICT	3	96.5	289.5
Entrepreneur	3	97.82	293.46
Business Math	4	99.3	397.2
Filipino	3	89.3	267.9
English	3	94.25	282.75
Statistics and Probability	4	92.75	371
TOTAL	Σf = 20	Σx = 569.92	Σf(x) = 1,901.81

Weighted Mean (W)

=

$\frac{\Sigma f(x)}{\Sigma f}$

Weighted Mean (W)

=

$\frac{1,901.81}{20}$

Weighted Mean (W)

=

95.0905 ≈ 95.09

c. Mean of Grouped Data

- The mean of grouped data may also be computed from a given frequency distribution.
- To compute the mean for grouped data, we can use two formulas, namely;

1. The Class Mark Formula

$\bar{x}$

=

$\frac{\Sigma fXm}{n}$

STEPS:

- Identify the class mark of each class interval.
- Multiply the class mark by the frequency (f) in each class
- Add the products obtained from step 2 and divide the sum by the total frequency (f).

2. The Coded Formula

$$\bar{x} = Xam + (\frac{\Sigma fd}{n})i$$

SAMPLE PROBLEM - Class Mark Formula

Consider the following frequency table summarizing the product sales in 20 days. Find the average daily sales.

Sales (Class interval)	Number of Days (f)	Class Mark (Xm)	Product (f · Xm)
₱13,001 - ₱15,000	1	14,000.50	14,000.50
₱11,001 - ₱13,000	2	12,000.50	24,0001.00
₱9,001 - ₱11,000	3	10,000.50	30,001.50
₱7,001 - ₱9,000	6	8,000.50	48,003.00
₱5,001 - ₱7,000	2	6,000.50	12,001.00
₱3,001 - ₱5,000	4	4,000.50	16,002
₱1,001 - ₱3,000	2	2,000.50	4,001.00
Class Size = 2,000	n = 20		ΣfXm = 148,010.00

$$\bar{x} = \frac{\Sigma fXm}{n} \quad \rightarrow \quad \bar{x} = \frac{\Sigma 148,010.00}{20} \quad \rightarrow \quad \bar{x} = 7,400.50$$

SAMPLE PROBLEM - Coded Formula

Sales (Class interval)	Number of Days (f)	Deviation (d)	Product (f · d)
₱13,001 - ₱15,000	1	3	3
₱11,001 - ₱13,000	2	2	4
₱9,001 - ₱11,000	3	1	3
₱7,001 - ₱9,000	6	0	0
₱5,001 - ₱7,000	2	-1	-2
₱3,001 - ₱5,000	4	-2	-8
₱1,001 - ₱3,000	2	-3	-6
Class Size = 2,000	n = 20		Σfd = -6

$$\bar{x} = Xam + (\frac{\Sigma fd}{n})i \quad \rightarrow \quad \bar{x} = 8,000.50 + (\frac{-6}{20})2000 \quad \rightarrow \quad \bar{x} = 7,400.50$$

B. MEDIAN

- Is the value in the midpoint or halfway mark data set, and divides the data set into two equal parts.
- Is the middle value of a given set of measurements. Provided that the values or measurements are arranged in an array. An array is an arrangement of values in increasing or decreasing order.
- Is the value that is located at the very center (or middle) that divides the data, meaning half of the observations fall above the median, and the other half fall below the median.

-> Characteristics of the MEDIAN

- The median is the most appropriate measure of central tendency for interval data.
- The median lies between the highest and the lowest measurements.
- There is only one value for the median in a given set of measurements.
- The median is not influenced by extreme values.
- The median is used when the middle value is desired. It is the value where 50% or half of the distribution lies above it and 50% lies below it.

a. Median for Ungrouped Data

STEPS:

1. Arrange the values in ascending or descending order.
 2. Locate the center value in the arranged data.
- If the number of observations in the data set is odd, then the median is the centrally located observation, that is, The Median is the $(\frac{n+1}{2})th$ term
 - If the number of observations in the data is even, then the median is the average of the two middle observations, that is, the median is the average of the $(\frac{n}{2})th$ and the $(\frac{n}{2} + 1)th$ term.

SAMPLE PROBLEM - Odd number of observations

	Sun	Mon	Tues	Wed	Thurs	Fri	Sat
Daily Rates	₱2,000	₱1,000	₱1,000	₱1,600	₱2,000	₱1,600	₱1,600

1. Arrange the values in ascending to descending order
2000, 2000, 1600, **1600**, 1600, 1000, 1000

SAMPLE PROBLEM - Even number of observations

25, 30, 32, 37, 40, 45, 48, 50

$\tilde{x} = \frac{37 + 40}{2} \qquad \tilde{x} = 38.5$

b. Median for Grouped Data

FORMULA:

$$\tilde{x} = LB + \left(\frac{(\frac{n}{2}) < cf}{fm}\right)i$$

SAMPLE PROBLEM

Sales (Class interval)	Number of Days (f)	<Cumulative Frequency
₱13,001 - ₱15,000	1	20
₱11,001 - ₱13,000	2	19
₱9,001 - ₱11,000	3	17
₱7,001 - ₱9,000	6	14
₱5,001 - ₱7,000	2	8
₱3,001 - ₱5,000	4	6
₱1,001 - ₱3,000	2	2
Class Size = 2,000	n = 20	

$$\tilde{x} = LB + (\frac{(\frac{n}{2}) < cf}{f_m})i \quad \rightarrow \tilde{x} = 7,000.5 + (\frac{(\frac{20}{2}) 8}{6})2000 \quad \rightarrow$$

$$\tilde{x} = 33,667.17$$

C. MODE

- It is the **value that appears most often** in a data set.
- It is obtained by selecting values that have the **highest frequency** in the data set.
- The value that **occurs most frequently** in a set of measurements or values.
- The mode for a particular set of data need not be unique. It may exist or not; **it is also possible to have two or more modes** in a single set of data.

-> Characteristics of the MODE

- The mode is the most appropriate measure of central tendency when the data are nominal in scale.
- The mode is the least reliable among the three measures of central tendency because its value is undefined in some distributions.
- The mode is used when we want to find the value that occurs more often.
- The mode is a quick approximation of the average. The mode is sometimes referred to as an inspection average.

a. Mode for Ungrouped Data

SAMPLE:
P100, P200, P500, P100, P300, P200, P100, P700 - Mode - 100, unimodal

b. Mode for Grouped Data

$$M_0 = l + (\frac{f_1 - f_0}{2f_1 - f_0 - f_2})h$$

Where:

- l = lower limit of the modal class
- h = size of the class interval (assuming all class sizes to be equal)
- f₁ = frequency of modal class
- f₀ = frequency of the class preceding(before) the modal class
- f₂ = frequency of the class succeeding(after) the modal class

SAMPLE PROBLEM

Class Interval	(f)
16 - 23	1
24 - 31	3
32 - 39	6
40 - 47	12
48 - 55	10
56 - 63	8
	n = 40

$$M_0 = l + (\frac{f_1-f_0}{2f_1-f_0-f_2})h$$

$$M_0 = 39.5 + (\frac{6}{8})8$$

$$M_0 = 39.5 + 6 = 45.5$$